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CANDIDATE – PLEASE NOTE!

You must sign below and return this booklet with the Answer Sheet. Failure to do so may result in disqualification.

____
Signature

FORM TP 2007104

TEST CODE **01234010**

JANUARY 2011

CARIBBEAN EXAMINATIONS COUNCIL

SECONDARY EDUCATION CERTIFICATE EXAMINATION

MATHEMATICS

Paper 01 – General Proficiency

90 minutes

04 JANUARY 2011 (p.m.)

READ THE FOLLOWING DIRECTIONS CAREFULLY

1. In addition to this test booklet, you should have an answer sheet.
2. Calculators and mathematical tables may NOT be used for this paper.
3. **A list of formulae is provided on page 2 of this booklet.**
4. This test consists of 60 items. You will have 90 minutes to answer them.
5. Each item in this test has four suggested answers, lettered (A), (B), (C), (D). Read each item you are about to answer, and decide which choice is best.
6. On your answer sheet, find the number which corresponds to your item and blacken the space having the same letter as the answer you have chosen. Look at the sample item below.

<u>Sample Item</u>

$2a + 6a =$

(A) $8a$
(B) $8a^2$
(C) $12a$
(D) $12a^2$

<u>Sample Answer</u>

● Ⓑ Ⓒ Ⓓ

The best answer to this item is "$8a$", so answer space (A) has been blackened.

7. If you want to change your answer, erase your old answer completely and fill in your new choice.
8. When you are told to begin, turn the page and work as quickly and as carefully as you can. If you cannot Answer an item, omit it and go on to the next one. You can return later to the item omitted. Your score will be the total number of correct answers.
9. You may do any rough work in the booklet.
10. Do not be concerned that the answer sheet provides spaces for more answers than there are items in this test.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

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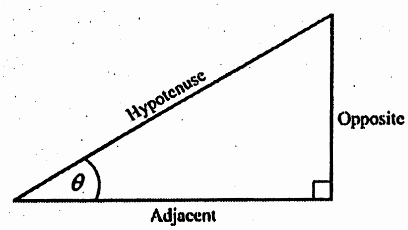
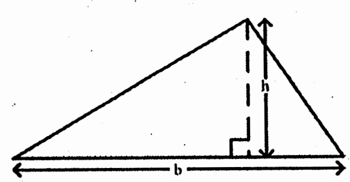
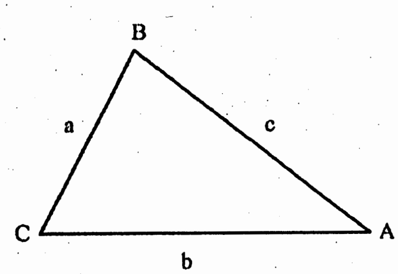
LIST OF FORMULAE

Volume of a prism	$V = Ah$ where A is the area of a cross-section and h is the perpendicular length.
Volume of a cylinder	$V = \pi r^2 h$ where r is the radius of the base and h is the perpendicular height.
Volume of a right pyramid	$V = \frac{1}{3}Ah$ where A is the area of the base and h is the perpendicular height.
Circumference	$C = 2\pi r$ where r is the radius of the circle.
Area of a circle	$A = \pi r^2$ where r is the radius of the circle.
Area of Trapezium	$A = \frac{1}{2}(a+b)h$ where a and b are the lengths of the parallel sides and h is the perpendicular distance between the parallel sides.
Roots of quadratic equations	If $ax^2 + bx + c = 0$, then $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
Trigonometric ratios	$\sin\theta = \frac{\text{opposite side}}{\text{hypotenuse}}$
	$\cos\theta = \frac{\text{adjacent side}}{\text{hypotenuse}}$
	$\tan\theta = \frac{\text{opposite side}}{\text{adjacent side}}$



Area of triangle	Area of $\Delta = \frac{1}{2}bh$ where b is the length of the base and h is the perpendicular height
	Area of $\Delta ABC = \frac{1}{2}ab\sin C$
	Area of $\Delta ABC = \sqrt{s(s-a)(s-b)(s-c)}$ where $s = \frac{a+b+c}{2}$
Sine rule	$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$
Cosine rule	$a^2 = b^2 + c^2 - 2bc\cos A$





1. What percentage of 30 is 6 ?

(A) 5%
(B) 18%
(C) 20%
(D) 150%

2. The number 3.14063 written correct to 3 decimal places is

(A) 3.140
(B) 3.141
(C) 3.146
(D) 3.150

3. Express $4\frac{3}{8}$ as a decimal correct to 3 significant figures.

(A) 4.30
(B) 4.37
(C) 4.38
(D) 4.40

4. There are 40 students in a class. Girls make up 60% of the class. 25% of the girls wear glasses. How many girls in the class wear glasses?

(A) 6
(B) 8
(C) 10
(D) 15

5. If \$560 is shared in the ratio $2:3:9$, the difference between the largest and the smallest shares is

(A) \$ 80
(B) \$240
(C) \$280
(D) \$360

6. If 60% of a number is 90, what is the number?

(A) 30
(B) 54
(C) 150
(D) 180

7. $\left(-\frac{1}{2}\right)^3$ is the same as

(A) $-\frac{1}{8}$
(B) $-\frac{1}{6}$
(C) $\frac{1}{8}$
(D) $\frac{1}{6}$

8. What is the highest Common Factor of the set of numbers $\{54, 72, 90\}$?

(A) 9
(B) 18
(C) 90
(D) 1080

9. If $3n$ is an odd number, which of the following is an even number?

(A) $3n-1$
(B) $3n-2$
(C) $3n+2$
(D) $3n+2n$

10. By the distributive law $49 \times 17 + 49 \times 3 =$

(A) $52 + 66$
(B) 52×66
(C) $49 + 20$
(D) 49×20

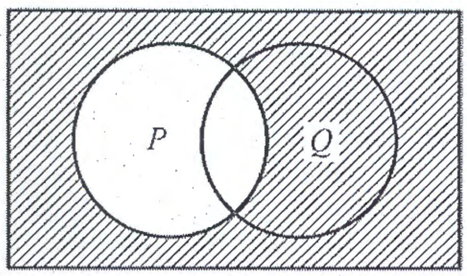
11. Which of the following sets is equivalent to $\{a, b, c, d\}$?

(A) $\{4\}$
(B) $\{a, b, c\}$
(C) $\{p, q, r, s\}$
(D) $\{1, 2, 3, 4, 5\}$

12. If $U = \{1,3,5,6,8\}$ and $A = \{3,6\}$, then the number of elements in A' is

(A) 2
(B) 3
(C) 4
(D) 8

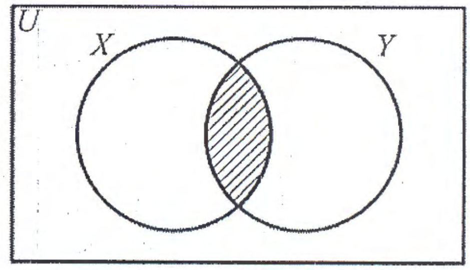
Item 13 refers to the Venn diagram below.



13. In the Venn diagram above, the shaded area represents

(A) P'
(B) $(P \cup Q)'$
(C) $P' \cap Q'$
(D) $P \cap Q'$

Item 14 refers to the Venn diagram below.



14. In the figure above, X represents the set of multiples of four. Y represents the set of multiples of 5. The shaded region represents the set of all multiples of

(A) 8
(B) 9
(C) 10
(D) 20

15. A plot of land is valued at $18000. Land tax is charged at a rate of $0.70 per $100. What is total amount of tax to be paid for the land?

(A) $110.00
(B) $126.00
(C) $180.70
(D) $257.15

16. A dress which costs $180 is being sold at a discount of 10%. The amount of the discount is

(A) $ 1.80
(B) $ 10.00
(C) $ 18.00
(D) $170.00

Item 17 refers to the table below.

House Insurance	50¢ per $100
Contents Insurance	25¢ per $100

17. The table above shows the rates charged by an insurance company. How much will a person pay for his insurance, if his house is valued at $50 000, and the contents at $10 000?

(A) $225
(B) $275
(C) $450
(D) $500

18. A man bought a cow for $200 and sold it to gain $50. What was his gain as a percentage of the cost price?

(A) 5%
(B) 15%
(C) 20%
(D) 25%

19. If the simple interest on $800 for 3 years is $54. What is the rate of interest per annum?

(A) $\frac{4}{9}\%$

(B) $2\frac{1}{4}\%$

(C) 5%

(D) 44%

20. A customer buys a table on hire purchase. He makes a deposit of $306 and pays six monthly installments of $60 each. The TOTAL cost to the customer is

(A) $360
(B) $366
(C) $666
(D) $966

21. A man pays 60 cents for every 200 m^3 of gas used, plus a fixed charge of $13.75. How much does he pay when he uses 55 000 m^3 of gas?

(A) $151.25
(B) $165.00
(C) $175.25
(D) $178.75

22. A company employs 12 gardeners at $26 per day, and 8 clerks at $17 per day. What is the mean daily wage, in dollars, of the 20 employees?

(A) $20.00
(B) $21.50
(C) $22.40
(D) $31.50

23. $\frac{4}{5x}+\frac{2}{5x}=$

(A) $\frac{6}{25x}$

(B) $\frac{8}{25x}$

(C) $\frac{6}{10x}$

(D) $\frac{6}{5x}$

24. If x is an integer that satisfies the inequality $4<2x\leq 6$, then

(A) $2<x\leq 3$
(B) $-2<x\leq 3$
(C) $-3<x\leq 2$
(D) $-3\leq x<-2$

25. The expression $-2(x-4)$ is the same as

(A) $-2x-8$
(B) $-2x-4$
(C) $2x+4$
(D) $-2x+8$

26. $3a(a+2b)-b(2a-3b)=$

(A) $3a^2-ab+3b^2$
(B) $3a^2+4ab+3b^2$
(C) $3a^2+4ab-3b^2$
(D) $3a^2+8ab-3b^2$

27. If $m*n=\sqrt{mn-n^2}$, then $5*3=$

(A) $\sqrt{6}$
(B) 3
(C) $\sqrt{15}$
(D) 6

28. John has x marbles and Max has twice as many. Max gives Tom 5 of his marbles. How many marbles does Max now have?

(A) $x+5$
(B) $x-5$
(C) $2x+5$
(D) $2x-5$

29. Given that $3(x-1)-2(x-1)=7$, then the value of x is

(A) 6
(B) 7
(C) 8
(D) 9

30. When 8 is subtracted from a number and the result is multiplied by 3, the final answer is 21. What is the **original** number?

(A) 1
(B) 3
(C) 10
(D) 15

31. The sum of two numbers x and y, is 18, and their difference is 14. Which pair of equations describes the above statement?

(A) $x+y=18$
$x-y=14$

(B) $x+y=32$
$x-y=4$

(C) $2x+2y=18$
$2x-2y=14$

(D) $xy=18$
$x-y=14$

32. The volume of a cube whose edge is 6 cm long is

(A) 18 cm^3
(B) 36 cm^3
(C) 72 cm^3
(D) 216 cm^3

33. 2500 millimetres expressed in metres is

(A) 0.25
(B) 2.5
(C) 25
(D) 250

34. The distance around the edge of a circular pond is 88 m. The radius, in metres is

(A) 88π

(B) 176π

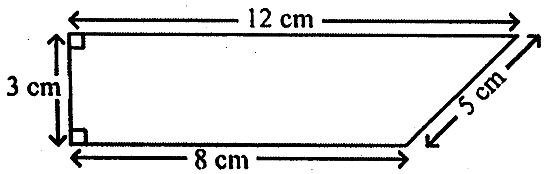
(C) $\dfrac{88}{\pi}$

(D) $\dfrac{88}{2\pi}$

35. A car travels 80 kilometres in $2\frac{1}{2}$ hours. What is its speed in kilometers per hour?

(A) 6
(B) 32
(C) 82.5
(D) 200

<u>Item 36</u> refers to the diagram below.



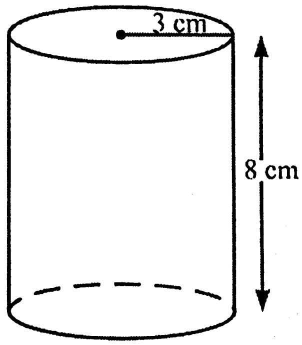
36. The area of the quadrilateral above is

(A) 24 cm^2
(B) 28 cm^2
(C) 30 cm^2
(D) 36 cm^2

37. The lengths of the sides of a triangle are x, $2x$ and $2x$ centimetres. If the perimeter is 20 centimetres, what is the value of x, in centimeters?

(A) 4
(B) 5
(C) 8
(D) 10

<u>Item 38</u> refers to the following diagram



38. The diagram above, **not drawn to scale,** shows a cylinder of radius 3 cm and height of 8 cm. The volume is.

(A) $12\pi\text{ cm}^3$

(B) $48\pi\text{ cm}^3$

(C) $72\pi\text{ cm}^3$

(D) $192\pi\text{ cm}^3$

39. A man leaves home at $22:15$ hrs and reaches his destination at $04:00$ hrs on the following day in the same time zone. How many hours did the journey take?

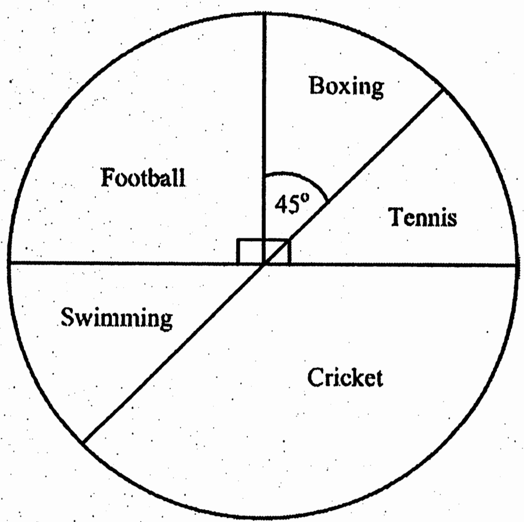
(A) 5

(B) $5\dfrac{3}{4}$

(C) 6

(D) $6\dfrac{1}{4}$

Items 40-42 refer to the diagram below which shows the sport chosen by 160 boys who participated in a games evening at their school



40. The number of boys who chose football is

(A) 40
(B) 90
(C) 110
(D) 150

41. How many boys participated in cricket?

(A) 54
(B) 60
(C) 110
(D) 120

42. The probability that a boy chosen at random participated in boxing is

(A) $\frac{1}{8}$

(B) $\frac{1}{4}$

(C) $\frac{1}{2}$

(D) $\frac{7}{8}$

43. The median of the numbers:
1, 1, 5, 5, 6, 7, 7, 7, 7, 8 is

(A) 5.4
(B) 6
(C) 6.5
(D) 7

44. A bag contains 2 red, 4 yellow and 6 blue balls. The probability of drawing a **blue** ball from the bag at random is

(A) $\frac{1}{2}$

(B) $\frac{1}{3}$

(C) $\frac{1}{6}$

(D) $\frac{6}{11}$

Item 45 refers to the following table.

Length of Leaf (cm)	10-14	15-19	20-24	25-29
Frequency	3	8	12	7

45. The lengths of 30 cabbage leaves were measured, to the nearest cm, and the information grouped as shown in the table above.

The class boundaries are

(A) 3,8,12,7
(B) 5,5,5,5
(C) 9.5,14.5,19.5,24.5,29.5
(D) 10,14,15,19,20,24,25,29

46. Which of the following line graphs represents $\{x : -2 < x \leq 4\}$?

(A) 

(B) 

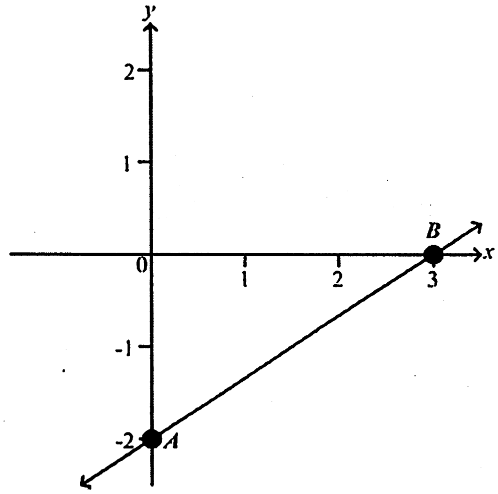
(C) 

(D) 

47. If $f(x) = x^2 - x - 1$, then $f(-5) =$

(A) -31
(B) 29
(C) 24
(D) 31

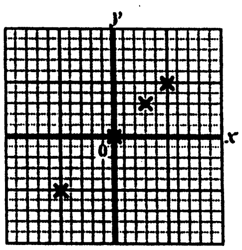
Item 48 refers to the graph below.

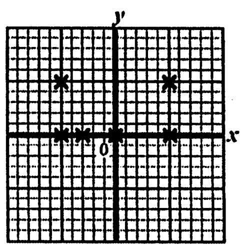


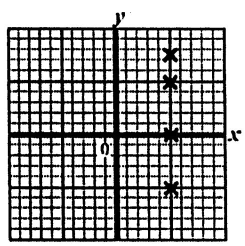
48. The straight line AB cuts the y axis at

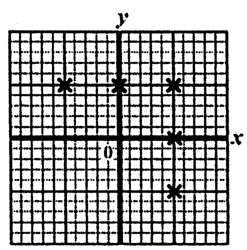
(A) $(0,3)$
(B) $(0,2)$
(C) $(3,-2)$
(D) $(0,-2)$

49. Which of the following represents the graph of a function?

I. 

II. 

III. 

IV. 

(A) I
(B) II
(C) III
(D) IV

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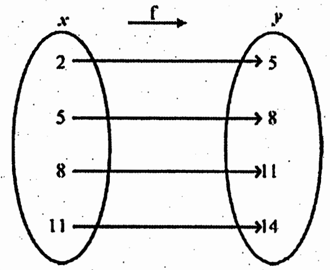
50.



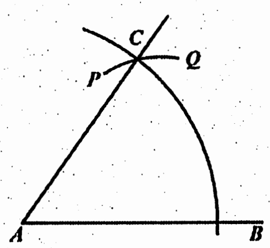
The arrow diagram above shows a function. Which of the following BEST describes the function?

(A) $f(x) = x + 3$
(B) $f(x) = y + 3$
(C) $x = y + 3$
(D) $y = x$

51. The range of $f : x \rightarrow x^3$ for the domain $\{-2, -1, 0, 1, 2\}$ is

(A) $\{0, 1, 8\}$
(B) $\{-2, -1, 0, 1, 2\}$
(C) $\{-6, -3, 0, 3, 6\}$
(D) $\{-8, -1, 0, 1, 8\}$

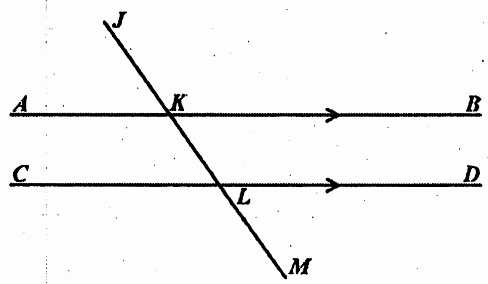
Item 52 refers to the diagram below of a construction. With centre A, an arc BC is drawn. With centre B, and the same radius, the arc PCQ is drawn.



52. What is the measure of $\angle BAC$?

(A) 30°
(B) 45°
(C) 60°
(D) 75°

Item 53 refers to the diagram below.

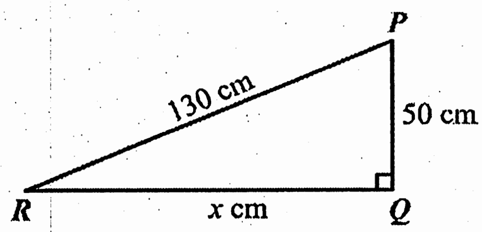


53. In the diagram, AB is parallel to CD, and $\angle JKB = 125°$.

$\angle MLD$ is

(A) 45°
(B) 55°
(C) 90°
(D) 125°

Item 54 refers to the following diagram.



54. In the right-angled triangle above, **not drawn to scale**, $\hat{Q} = 90°$, $PQ = 50$ cm, $PR = 130$ cm and $RQ = x$ cm.

Tan $P\hat{R}Q =$

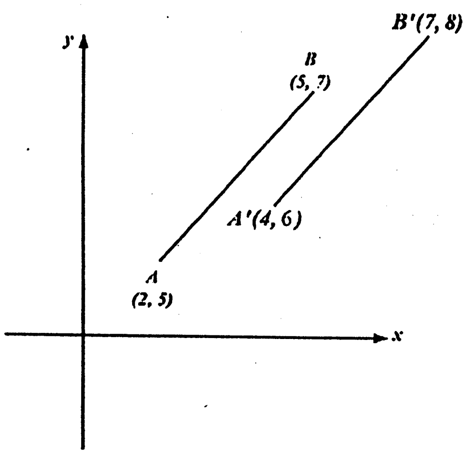
(A) $\frac{50}{x}$

(B) $\frac{x}{50}$

(C) $\frac{50}{130}$

(D) $\frac{x}{130}$

Item 55 refers to the diagram below.



55. In the diagram above, the translation by which AB is mapped onto $A'B'$ is represented by

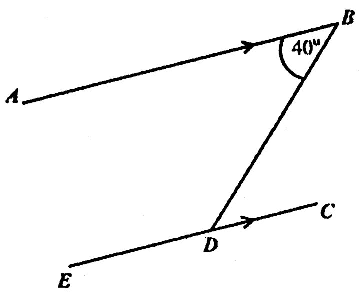
(A) $\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

(B) $\begin{pmatrix} 2 \\ 1 \end{pmatrix}$

(C) $\begin{pmatrix} 3 \\ 2 \end{pmatrix}$

(D) $\begin{pmatrix} 5 \\ 3 \end{pmatrix}$

Item 56 refers to the diagram below.

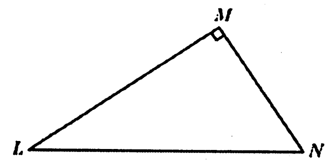


56. AB is parallel to EC. Calculate $\angle BDE$.

(A) 40°
(B) 50°
(C) 140°
(D) 180°

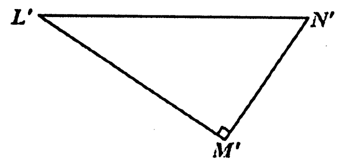
Item 57 refers to the diagram below.

57.

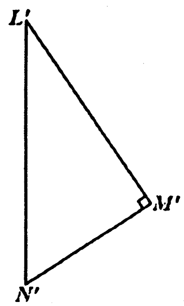


The triangle LMN is rotated in a clockwise direction about L through an angle of 90°. What is its image?

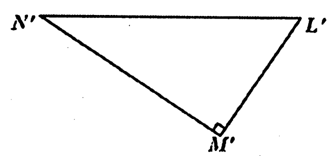
(A)



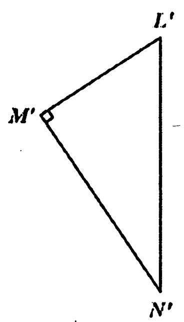
(B)



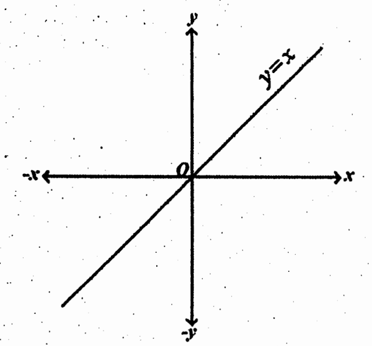
(C)



(D)



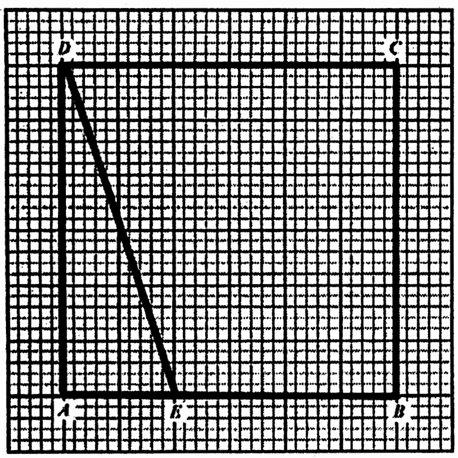
Item 58 refers to the following diagram.



58. In the diagram above, if the line $y = x$ is rotated anti-clockwise about O through 90^o, what is its image?

(A) $y = 0$
(B) $x = 0$
(C) $y = x$
(D) $y = -x$

Item 59 refers to the following diagram.

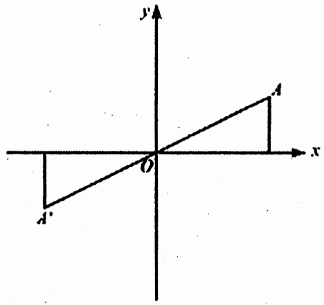


59. How many triangles congruent to ΔADE would be needed to cover the rectangle $ABCD$ entirely?

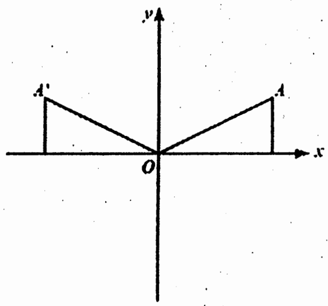
(A) 8
(B) 6
(C) 4
(D) 2

60. In each of the diagrams shown below, A' is the image of A. Which of the following diagrams shows a reflection in the x axis ?

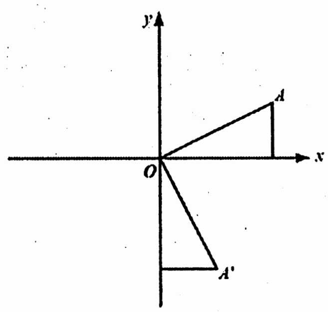
(A)



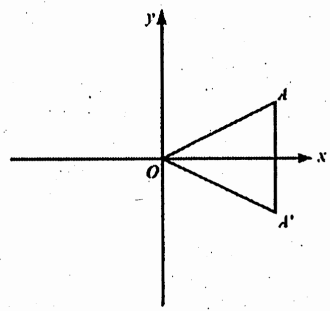
(B)



(C)



(D)



IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.